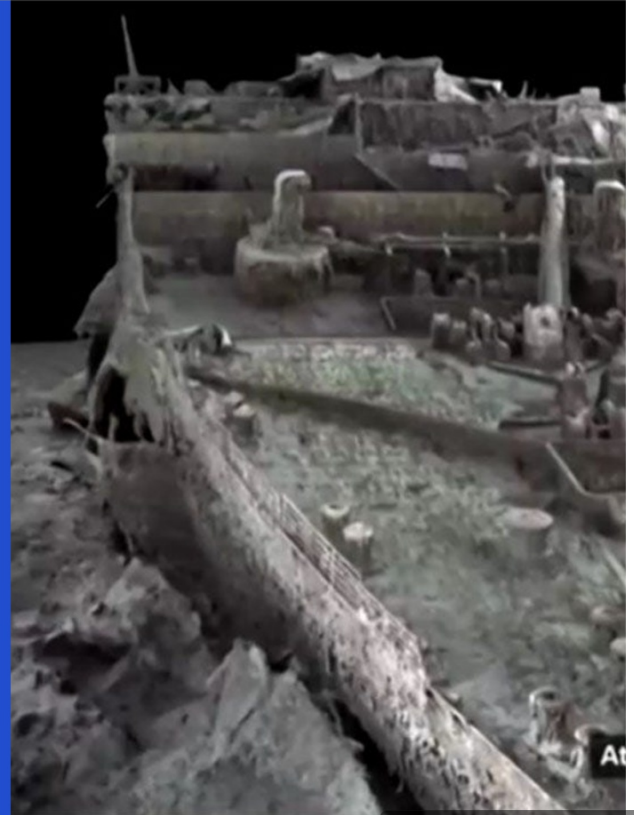


Data Science Project Brief – Predicting Survival of Titanic Passengers

A comprehensive analysis of a data science project focused on Titanic passenger survival predictions.

P

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Presenter



Overview of the Titanic Dataset

Data Science Project Code Brief - Predicting Survival of Titanic Passengers

01

PassengerId

Unique identifier for each passenger, facilitating tracking and analysis.

02

Survived

Indicates survival status (0 = No, 1 = Yes), crucial for prediction modeling.

03

Pclass

Ticket class (1, 2, or 3), providing insight into socioeconomic status.

04

Demographic Information

Includes Name, Sex, and Age, essential for understanding passenger profiles.

05

SibSp

Number of siblings/spouses aboard, indicating family connections.

06

Parch

Number of parents/children aboard, further detailing family structures.

07

Ticket

Ticket number assigned to each passenger, useful for tracking.

08

Fare

Amount paid for the ticket, which may correlate with survival rates.

09

Cabin

Cabin number indicating the location of passengers on the ship.

Data Import and Preprocessing

Key steps for preparing Titanic passenger dataset for analysis



Importing Libraries

Utilize libraries like NumPy and Pandas for data manipulation and analysis.



Loading Datasets

Read the training and testing datasets from CSV files using Pandas.



Combining Datasets

Concatenate training and testing datasets to ensure uniform preprocessing.



Null Value Exploration

Identify and analyze null entries in critical columns like 'Age' and 'Cabin'.

Exploratory Data Analysis (EDA)

Insights on Passenger Age and Fare in Titanic Dataset

29.7

Mean Age of Passengers

The average age of Titanic passengers is approximately 29.7 years, indicating a relatively young demographic.

32.2

Average Fare Paid

The average fare paid by passengers was around \$32.20, reflecting the economic diversity among the travelers.

Categorical Data Analysis

Analyzing Survival Rates Based on Gender and Ticket Class

Sex	Pclass
Women	Higher Survival Rate
Men	Lower Survival Rate
1st Class	Better Survival Rate
2nd Class	Moderate Survival Rate
3rd Class	Lower Survival Rate

Correlation Analysis

Investigating relationships between numeric features using a heatmap.

Numeric Features	Correlation Coefficient
Age and Fare	Weak Positive Correlation
SibSp and Parch	Moderate Correlation

Created using  presentations

Feature Engineering: Cabin and Ticket

Enhancing Features for Titanic Survival Prediction



Cabin Analysis

Create a new feature 'cabin_multiple' to capture the number of cabins associated with each passenger.



Ticket Classification

Classify tickets into numeric and non-numeric categories to streamline data processing.



New Feature – cabin_multiple

'cabin_multiple' represents the count of cabins linked to each passenger, providing depth to the dataset.



New Feature – numeric_ticket

'numeric_ticket' serves as an indicator showing whether a ticket is numeric, enhancing data usability.



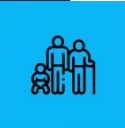
Data Cleaning and Imputation

Essential steps for preparing data in Titanic survival prediction



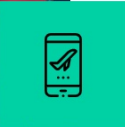
01 Dropping Null Values

Remove nulls from the 'Embarked' column, which only has 2 missing entries.



02 Imputing Age Values

Fill missing Age values using the mean of the Age column from the training dataset.



03 Imputing Fare Values

Fill missing Fare values with the mean Fare from the training dataset to ensure data completeness.

Normalization and Scaling

- 01 Importance of Scaling**
Scaling helps to ensure that features contribute equally to the distance calculations in algorithms.
-

- 02 StandardScaler Usage**
StandardScaler normalizes features by removing the mean and scaling to unit variance.
-

- 03 Fit and Transform**
Use fit_transform on training data and transform on test data to maintain consistency.

Model Selection: Initial Candidates

Evaluating Models for Predicting Titanic Passenger Survival



01 Gaussian Naïve Bayes

A probabilistic model that applies Bayes' theorem with strong independence assumptions between features.



02 Logistic Regression

A statistical method for predicting binary classes using a logistic function.



03 Decision Tree

A model that uses a tree-like graph to make decisions based on feature splits.



04 K-Nearest Neighbors

A non-parametric method used for classification by comparing the closest training examples.



05 Random Forest

An ensemble method that uses multiple decision trees to improve classification accuracy.



06 Support Vector Classifier

A model that finds the optimal hyperplane for classification in high-dimensional spaces.

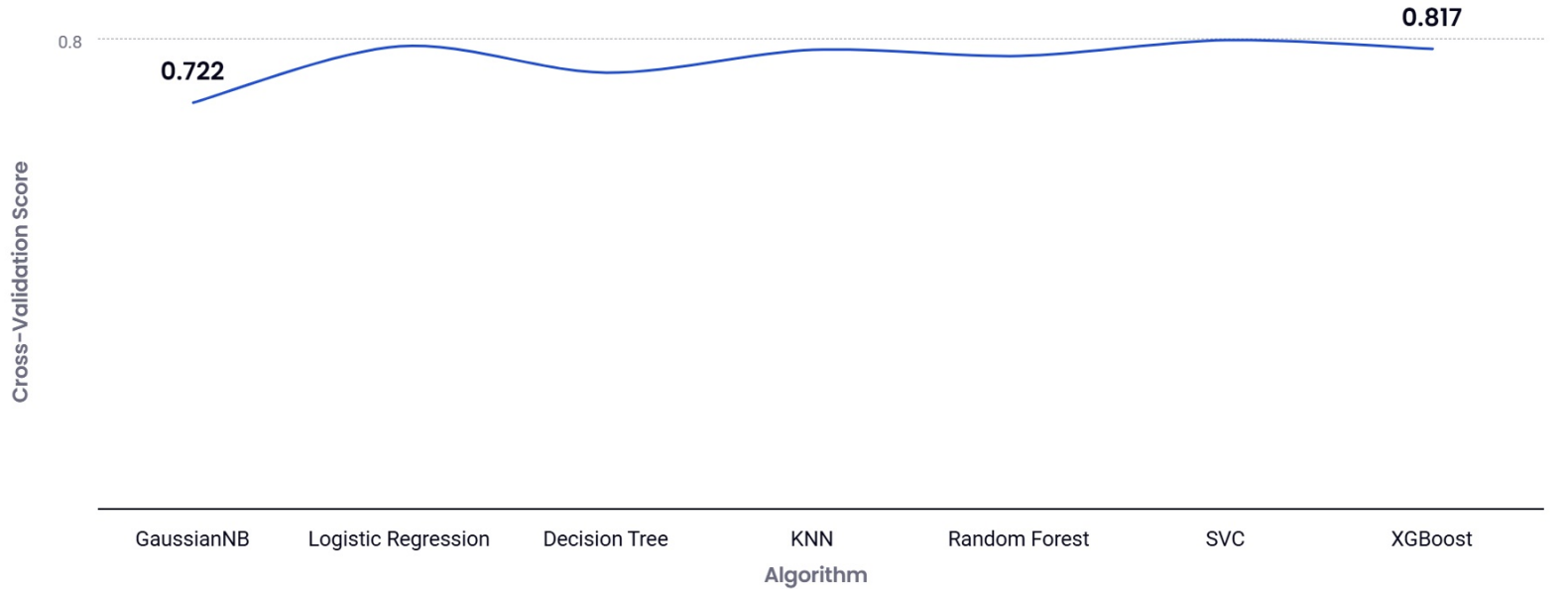


07 XGBoost

An optimized gradient boosting framework that is highly efficient and flexible.

Cross-Validation Scores

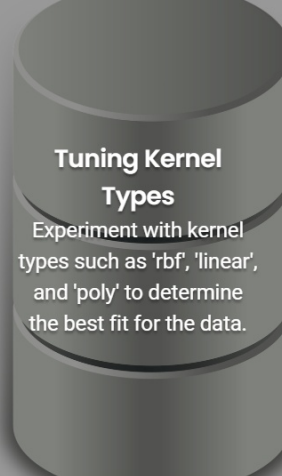
Performance Metrics of Various Classification Algorithms



Source: Companies Market Cap

Hyperparameter Tuning

Optimizing SVC Parameters for Enhanced Model Performance



Gamma Adjustment for 'rbf'
Fine-tune the gamma parameter for the 'rbf' kernel to influence the decision boundary's flexibility.

A graphic of a vertical bar with a color gradient from blue at the bottom to yellow at the top, representing the process of adjusting the gamma parameter.



Final Predictions and Submission

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Generate predictions on test data

Utilize the best model to generate predictions on the scaled test dataset.



Export results for submission

Store the predictions in a dataframe and export the results to a CSV file for final submission.



Project Conclusion

Key insights and future directions for survival prediction



Importance of Data Preprocessing

Data preprocessing and feature engineering are crucial for model accuracy and effectiveness.



Model Performance

Support Vector Classifier (SVC) outperformed other models in predicting survival rates.



Socio-Economic Factors

The effect of socio-economic factors on survival is significant and should be further analyzed.



Future Work Directions

Future work should explore more features and advanced modeling techniques to enhance predictions.

References and Resources

Essential tools and materials for Data Science projects



01 Titanic Dataset from Kaggle

A rich dataset that serves as an excellent case study for predictive modeling.



02 Scikit-learn documentation

Comprehensive guides for implementing various Machine Learning models effectively.



03 Seaborn and Matplotlib

Powerful libraries for creating informative visualizations to analyze data trends.



04 Data Science projects with real-world datasets

Encourages practical application of theories to solve real-world problems.



05 Model evaluation techniques

Learning effective ways to assess model performance and accuracy in Machine Learning.

Explore Data Science Applications

Learn about predictive modeling through Titanic survival data.

